MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143E1 MECHANICAL PARTS AND ASSEMBLY FOR

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Quality Assurance Provisions and special requirements for the fabrication of the Mechanical Assembly for the PATRIOT M143E1 Safety and Arming Device.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Non-leaded; Flat Products (Plate, Bar, Sheet and Strip)
QQ-S-700 - Steel Sheet and Strip, Medium and High Carbon

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-S-48687 - Safety and Arming Device, GM: M143E1 Loading, Assembling and Packing For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1336
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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-810	- Environmental Test Methods
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9390590	- Mechanical Assembly

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A108	-Steel Bars, Carbon, Cold-Finished, Standard Quality, Specification for
ASTM A167	-Stainless and Heat Resisting Chromium Steel Plate, Sheet and Strip
ASTM A581	-Free Machining Stainless and Heat Resisting Steel Wire, Specification for
ASTM B194	-Copper-Berryllium Alloy Plate, Sheet, Strip and Rolled Bar, Specification for
ASTM B487	-Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Method for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall uspersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications and standards.

3.1.2 Polymer coating (12630866). The parts requiring coating (12630866), shall be coated as indicated on their respective drawings.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing (Dwg) 9390590 and associated drawings, specifications, and standards.

3.2.1 "G" Weight assembly (9752414) Bushing shall be staked and secured to flush or below flush to "G" weight. Stake shall withstand an axial load of 10 lbs., min without deforming bushing hole.

3.2.2 Escapement assembly (12630880).

3.2.2.1 Top plate (12630882). The pillars and plate shall withstand a push-off force of 25+2 lbs when applied in the direction indicated on drawing (12630880).

3.2.2.2 Bottom plate (12630881). The pillar and plate shall withstand a push-off force of 15+2 lbs when applied in the direction indicated on drawing (12630880).

3.2.2.3 Lever assembly (12630894).

3.2.2.3.1 Lever shaft (12630897). The shaft and lever shall withstand a push-off force of 15+2 lbs when applied in the direction indicated on drawing (12630894).

3.2.2.3.2 Lever pallet (12630196). The pallet and lever shall withstand a push-off force of 5+1 lbs when applied in the direction indicated on drawing (12630894).

3.2.2.4 Gear-Pinion assemblies (12630891, 12630888, 12630885). The assemblies shall withstand a push-off force as specified on their respective drawings.

3.3 Mechanical assembly (9390590).

3.3.1 Continuity. The electrical resistances shall meet the requirements specified in Table I for the positions indicated:

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TABLE I. Resistance

Identification	Designation	Terminals	Resistance (Ohms) Safe & Locked	Safe & UnLocked	Armed & Locked
Rotor Monitor (Left)	S2 (Left)	E1-E2	*GT 20M	GT 20M	1.0 Max
Rotor Monitor (Right)	S2 (Right)	E1-E2	GT 20M	GT 20M	1.0 Max
Latch Monitor (Left)	S1 (Left)	E1-E3	1.0 Max	GT 20M	1.0 Max
Latch Monitor (Left)	S1 (Left)	E1-E2	GT 20M	1.0 Max	GT 20M
Latch Monitor (Right)	S1 (Right)	E1-E3	1.0 Max	GT 20M	1.0 Max
Latch Monitor (Right)	S1 (Right)	E1-E2	GT 20M	1.0 Max	GT 20M
Condition Monitor (Left)	S2 (Left)	E4-E3	1.0 Max	1.0 Max	GT 20M
Condition Monitor (Right)	S2 (Right)	E4-E3	1.0 Max	1.0 Max	GT 20M
Detonator Contacts (Left)	S2 (Left)	E5-E6	GT 20M	GT 20M	1.0 Max
Detonator Contacts (Right)	S2 (Right)	E5-E6	GT 20M	GT 20M	1.0 Max
Solenoid (Left)	Left	Across Leads	12.3 to 14.9	Not Required	Not Required
Solenoid (Right)	Right	Across Leads	12.3 to 14.9	Not Required	Not Required

*Greater than (GT)

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3.3.2 Isolation resistance. The isolation resistance shall not be less than 20 megohms when measured between the base plate and the locations (See figure 1) indicated in both positions (not at the same time) listed in Table II.

TABLE II. Isolation resistance

	Position	
	Safe & Locked	Armed & Locked
a.	S1 (R) E1	S1 (R) E1
b.	S1 (R) E2	S1 (R) E2
c.	S2 (R) E1	S2 (R) E1
d.	S2 (R) E2	S2 (R) E3
e.	S2 (R) E4	S2 (R) E4
f.	S2 (R) E5	S2 (R) E5
g.	S2 (R) E6	S2 (R) E6
h.	Solenoid (R) Lead*	N/A
i.	S1 (L) E1	S1 (L) E1
j.	S1 (L) E2	S1 (L) E2
k.	S2 (L) E1	S2 (L) E1
l.	S2 (L) E2	S2 (L) E3
m.	S2 (L) E4	S2 (L) E4
n.	S2 (L) E5	S2 (L) E5
o.	S2 (L) E6	S2 (L) E6
p.	Solenoid (L) Lead*	N/A

*Only one lead wire is required.

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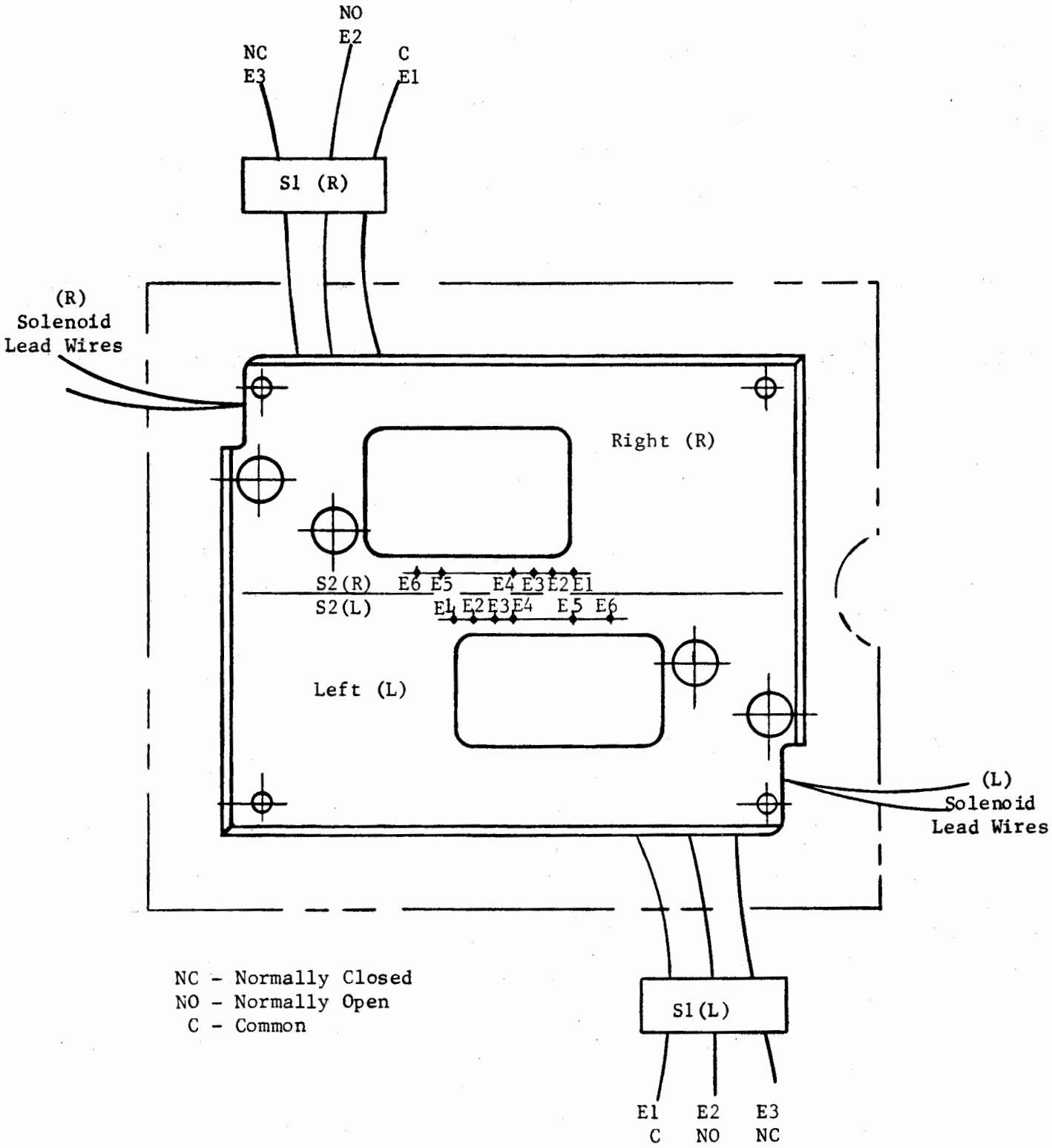


Figure 1

Top view (mechanical assembly)

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3.3.3 Intent-to-launch (ITL). With the Mechanical Assembly in the Safe and Locked position, the resistance measured between Latch Monitor (Right) S1 (Right) E1 to E3 and Latch Monitor (Left) S1 (Left) E1 to E3 shall be 1.0 ohm maximum. The application of a 25.2 - 1.0V pulse for a duration of 180 ± 10 milliseconds to both sets of solenoid leads connected in parallel shall unlock the Mechanical Assembly. This unlocking shall be indicated by a resistance of 1.0 ohm (max) when measured between the Latch Monitor (Right) S1 (Right) E1 to E2 and Latch Monitor (Left) S1 (Left) E1 to E2. This measurement shall be taken 215 ± 15 milliseconds after the start of the applied pulse.

3.3.4 Detonator rotor arming. Arming is defined as the detonator rotors rotating from the Safe and Unlocked position (See Table I Rotor Monitor Right and Rotor Monitor Left) to the Armed position (See Table I Rotor Monitor Right and Rotor Monitor Left). The Mechanical Assembly shall be Locked in the Armed position (See Table I Latch Monitor Right and Latch Monitor Left) ± 300 milliseconds of arming.

3.3.4.1 Detonator rotor arming time. With the Mechanical Assembly in the Safe and Unlocked position, each detonator rotor shall arm in not less than 3.1 seconds and not greater than 4.2 second when the -Z axis (See Figure 3) is subjected to 11.9 ± 0.2g's at a temperature of 70 ± 18°F.

3.3.5 Non-arming. With the Mechanical Assembly in the Safe and Unlocked position (See Table I), neither detonator rotor shall rotate to the Armed and Locked position (See Table I when subjected to a constant acceleration of 4.0 ± 0.5g in the -Z axis for a period of 20 seconds min (See Figure 2 & 3).

3.3.6 Return to safe. With no simulated g force applied, the Mechanical Assembly shall return to the Safe and Locked position (See Table I Safe & Locked Condition) from the Armed and Locked position (See Table I Armed & Locked Condition) when a Return to Safe Pulse of 35.0 ± 1.0V having a duration of 350 ± 150 milliseconds is applied to both solenoids simultaneously (See Figure 1). The Mechanical Assembly may be rearmed and resafed or removed from the test fixture (if used) and placed base down and subjected to additional return to safe pulses or both. The frequency of application of pulses shall not exceed a rate of four (4) pulses per minute.

3.3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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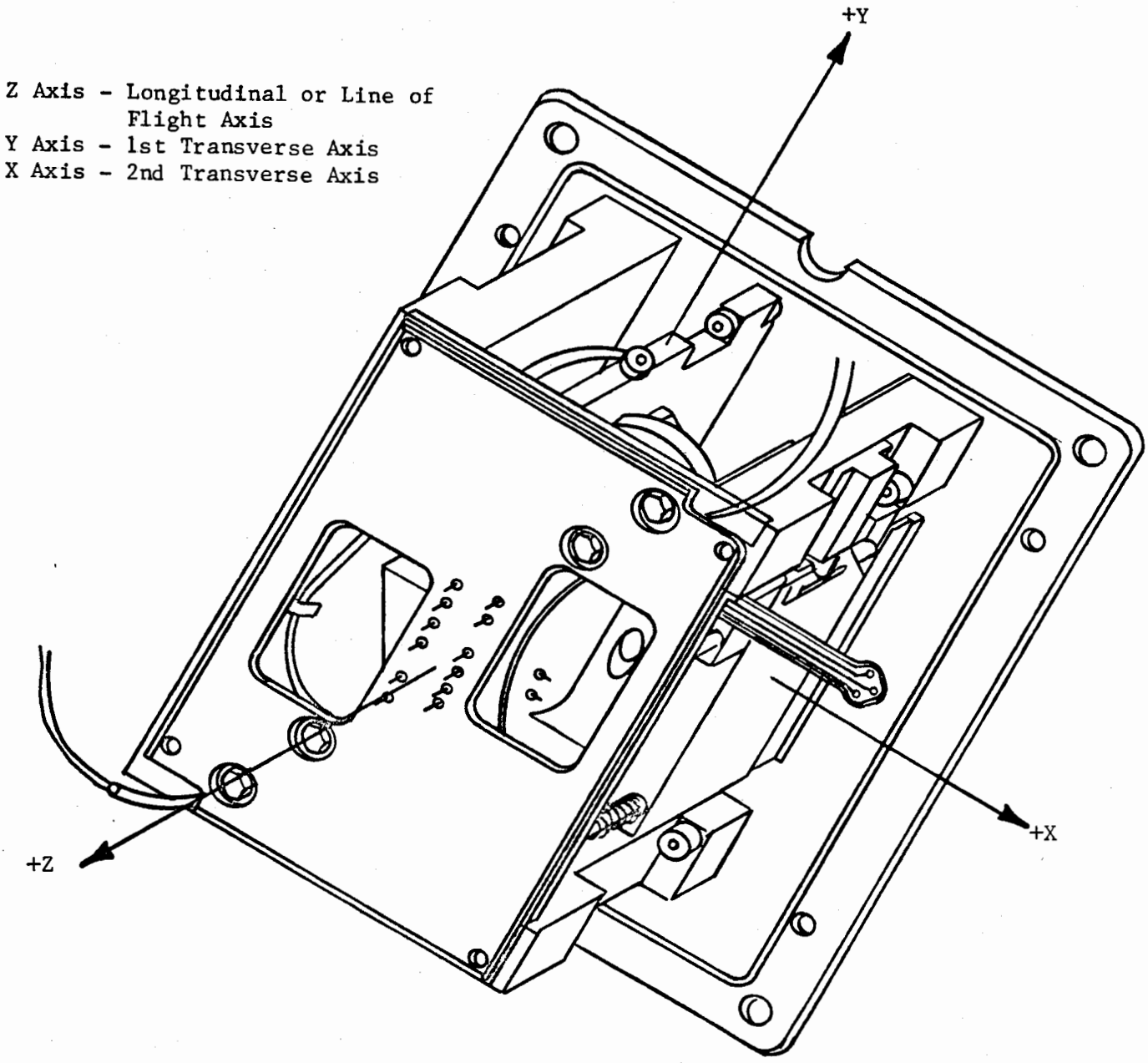


Figure 2

Mechanical assembly orientation

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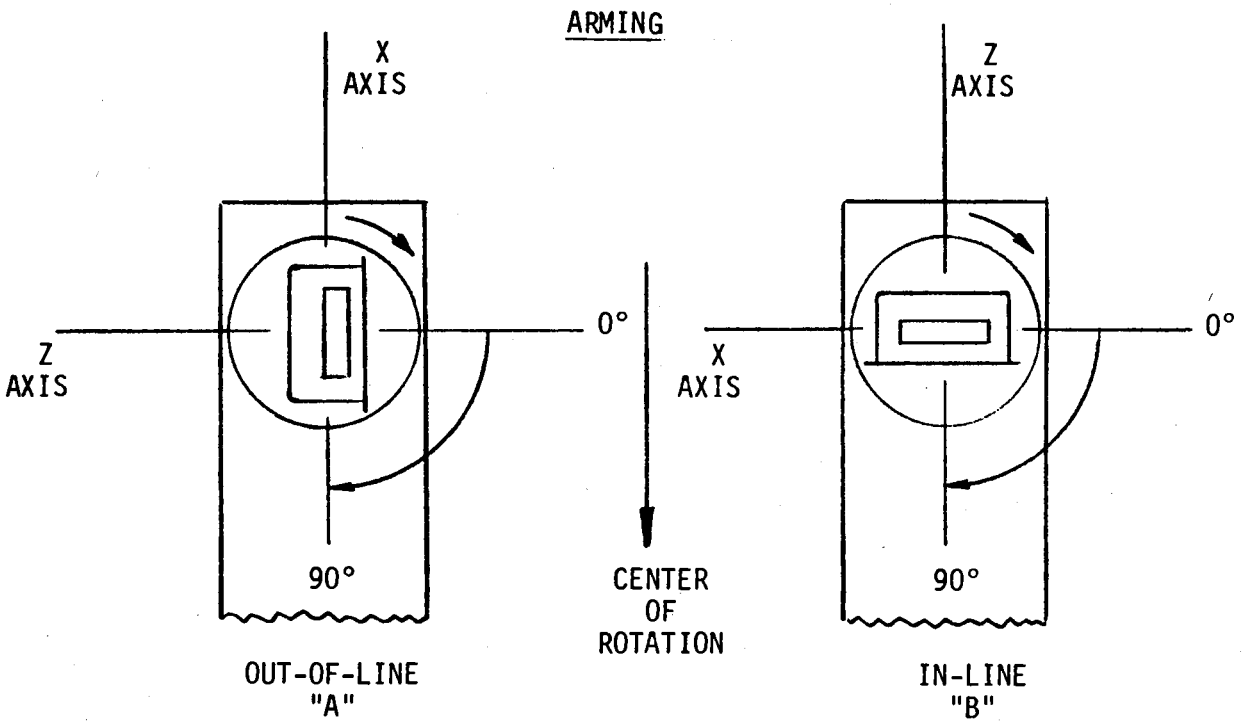


Figure 3 a & b

Patriot S&A orientation on centrifuge arm

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3.4 Workmanship.

3.4.1 General. All parts and assemblies shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, unblended radii, cold solder joints, surface defects, cracks, grease, dirt, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.4.2 Molded parts. The molded parts shall be free of porosity, warpage in excess of drawing dimensions, checks, cracks, chipped edges, scratches and other defects which would affect their serviceability.

3.4.3 Adhesives and lubricants. Adhesives or lubricants shall not be used in the mechanical assembly if the shelf life has expired or if the life is reduced due to accelerated aging.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First article inspection

4.3.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Drawings	Part Description	Quantity
12630890	Pinion, Number 2	5
12630889	Gear, Number 2	5
12630888	No. 2 Gear-Pinion Assembly	5
12630893	Pinion, Escapement	5
12630892	Gear, Escapement	5
12630891	Escapement Gear-Pinion Assembly	5
12630887	Pinion, No. 1	5
12630886	Gear, No. 1	5
12630885	No. 1 Gear-Pinion Assembly	5
12630897	Shaft, Lever	5
12630896	Pallet, Lever	5
12630895	Lever	5
12630894	Lever Assembly	5
12630881	Plate, Bottom	5
12630882	Plate, Top	5
12630883	Pillar, Number 1	5
12630884	Pillar, Number 2	5
12630880	Escapement Assembly	5
12630824	Contact, Electrical	5
12630823	Board, Switch	5

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12630822	Switch Board Assembly	5
9390601	Plate, Center	5
9390600	Center Plate Assembly	5
9390612	Housing, "G" Weight Latch	5
12630814	Latch, "G" Weight	5
12630815	Pin "G" Weight Latch	5
12630816	Spring, "G" Weight Latch	5
12630813	"G" Weight Latch Housing Assembly	5
12630806	Plate, Shorting Detonator	5
12630805	Pin, Terminal	5
12630804	Shorting Plate Detonator Assembly	5
12630803	Resistor Shorting Plate Detonator Assembly	5
12630807	Housing, Detonator	5
12630802	Resistor Shorting Plate Housing Detonator Assy	5
9390589	Spring, "G" Weight	5
9392412	"G" Weight	5
9392413	Bushing	5
9392414	"G" Weight Assembly	5
12630811	Weight, Rotor	5
12630812	Plate, Shorting (Safe & Arm)	5
9392415	Rotor, Detonator	5
12630810	Detonator Rotor Subassembly	5
9390610	Actuator, Latch	5
12630826	Solenoid, Rotary	5

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9390608	Latch Actuator Solenoid Assembly	5
9390607	Plate, Top	5
9390606	Top Plate Assembly	5
9390603	Tee Spacer	5
9390602	Tee Assembly	5
9390604	Shafting	5
9390605	Rail	5
9390596	Switch, Lever	5
9390595	Lever Switch Assembly	5
9390599	Pin, Rotor Latch	5
9390598	Latch, Rotor	5
9390597	Rotor Latch Assembly	5
9390593	Detent, Rotor Latch	5
12630820	Bearing, Rotor Latch Detent	5
12630817	Anchor, Rotor Latch Spring	5
12630818	Spring, Rotor Latch Detent	5
12630819	Spring, Rotor Latch	5
9390594	Plate, Outer	5
12630847	Adjuster Nut, Switch	5
9390592	Outer Plate Assembly	5
9390613	Screw, Self Sealing	5
9390611	Board Stiffener	5
9390591	Plate, Base	5
9390615	Gear, Escapement	5
9390616	Rotor, Balance	5
9392416	Balance Rotor Assembly	5
9390590	Mechanical Assembly	5

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4.3.1.1 Homogeneity. The first article shall comply with the homogeneity criteria of MIL-A-48078, 4.4.1.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

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TABLE III. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.3.4	TITLE Safety and Arming Device, GM: M143E1, Mechanical Assembly, Components and Subassemblies		SHEET 1 OF 1		DRAWING NUMBER See 4.3.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Mechanical Assembly, Components and Subassemblies (Drawing list - paragraph 4.3.1) Examination for defects	5		3	4.4.2.1 - 4.4.2.71

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630890
4.4.2.1	Pinion, Number 2				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pinion profile		0.40%	3.2	Gage
102	Coating, Thickness	1	Coupon	3.1.2	Gage/4.5.6.3
103	Coating, Adhesion	1	Coupon	3.1.2	Test4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630889
4.4.2.2	Gear, Number 2				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gear profile		0.40%	3.2	Gage
102	Coating, Thickness	1	Coupon	3.1.2	Gage/4.5.6.3
103	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630888
4.4.2.3	No. 2 Gear-Pinion Assembly				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Static axial load, non destructive		0.40%	3.2.3	Test
102	Perpendicularity (after axial load)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630893
4.4.2.4	Pinion, Escapement				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pinion profile		0.40%	3.2	Gage
102	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
103	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12630892
4.4.2.5	Gear, Escapement		NEXT HIGHER ASSEMBLY 12630880

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of Outside Diameter to -A-		0.40%	3.2	Gage
102	Gear profile		0.40%	3.2	Gage
103	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
104	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630891
4.4.2.6	Escapement Gear-Pinion Assembly				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Static axial load (Non destructive)		0.40%	3.2.3	Test
102	Perpendicularity (after axial load)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Pinion, No. 1		12630887
			NEXT HIGHER ASSEMBLY 12630880

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pinion profile		0.40%	3.2	Gage
102	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
103	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630886
4.4.2.8	Gear, No. 1				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gear profile		0.40%	3.2	Gage
102	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
103	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630885
4.4.2.9	No. 1 Gear-Pinion Assembly				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Static axial load (non destructive)		0.40%	3.2.3	Test
102	Perpendicularity (after axial load)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Shaft, Lever		12630897
			NEXT HIGHER ASSEMBLY
			12630880

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of pivots		0.40%	3.2	Gage
102	Shoulder length		0.40%	3.2	Gage
103	Pivot finish (2 places), before plating		0.40%	3.2	Gage
104	Pivot shoulder finish (2 places)		0.40%	3.2	Gage
105	Taper diameter runout		0.40%	3.2	Gage
106	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
107	Coating, adhesion	1	Coupon	3.1.2	Gage 4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630896
4.4.2.11	Pallet, Lever				NEXT HIGHER ASSEMBLY 12630894
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter, large		0.40%	3.2	Gage
102	Length, min.		0.40%	3.2	Gage
103	Finish		0.40%	3.2	Gage
104	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
105	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630895
4.4.2.12	Lever				NEXT HIGHER ASSEMBLY 12630894
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of shaft hole (max)		0.40%	3.2	Gage
102	Diameter of pallet pin hole (max) (2 places)		0.40%	3.2	Gage
103	Location of pallet pin holes (2 places)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630894
4.4.2.13	Lever Assembly				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Stake missing		0.40%	3.2	Visual
102	Lever pallet push-off (non destructive)		0.40%	3.2.2.3.2	Test
103	Shaft push-off (non destructive)		0.40%	3.2.2.3.1	Test
104	Lever pallet protrusion, max.		0.40%	3.2.2.3	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Plate, Bottom		12630881
			NEXT HIGHER ASSEMBLY 12630880

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of #2 Gear & Pinion (G&P) hole		0.40%	3.2	Gage
102	Location of #2 (G&P) hole		0.400%	3.2	Gage
103	Diameter of #1 pivot hole (3 places)		0.40%	3.2	Gage
104	Location of #1 pivot hole (3 places)		0.40%	3.2	Gage
105	Diameter of pillar hole (max) (2 places)		0.40%	3.2	Gage
106	Location of pillar hole max (2 places)		0.40%	3.2	Gage
107	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
108	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Gage
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630882
4.4.2.15	Plate, Top				NEXT HIGHER ASSEMBLY 12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of #2 Gear & Pinion (G&P) pivot hole		0.40%	3.2	Gage
102	Location of #2 G&P pivot hole		0.40%	3.2	Gage
103	Diameter of #1 pivot hole (3 places)		0.40%	3.2	Gage
104	Location of #1 pivot hole (3 places)		0.40%	3.2	Gage
105	Diameter of pillar holes (max) (3 places)		0.40%	3.2	Gage
106	Location of pillar holes (3 places)		0.40%	3.2	Gage
107	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
108	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630883
4.4.2.16	Pillar, Number 1				NEXT HIGHER ASSEMBLY 12670880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter, large end (min.)		0.40%	3.2	Gage
102	Diameter, small end (min.)		0.40%	3.2	Gage
103	Length, small diameter end (min.)		0.40%	3.2	Gage
104	Length, shoulder		0.40%	3.2	Gage
Minor					
210	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Pillar, Number 2				12630884
					NEXT HIGHER ASSEMBLY
					12630880
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter, large end, min.		0.40%	3.2	Gage
102	Diameter, small end, min.		0.40%	3.2	Gage
103	Length, large diameter end		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630880
4.4.2.18	Escapement Assembly				NEXT HIGHER ASSEMBLY 9390600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Stake missing, pillar above flush, Pillar #1 (bottom plate)		0.40%	3.2	Visual
102	Stake missing, Pillar #2 (2 places, top plate)		0.40%	3.2	Visual
103	Stake missing, Pillar #2 (2 places, bottom plate)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630824
4.4.2.19	Contact, Electrical				NEXT HIGHER ASSEMBLY 12630822
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height of contact, free position		0.40%	3.2	Gage
102	Length of contact, free position		0.40%	3.2	Gage
103	Radius of contact surface		0.40%	3.2	Gage
104	Finish of contact surface		0.40%	3.2	Gage
105	Thickness		0.40%	3.2	Gage
Minor					
201	Length of contact, short length		0.65%	3.2	Gage
202	Width of contact		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Board, Switch				12630823
					NEXT HIGHER ASSEMBLY 12630822
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Location of locating hole		0.65%	3.2	Gage
202	Diameter of locating hole		0.65%	3.2	Gage
203	Width of locating slot		0.65%	3.2	Gage
204	Location of locating slot		0.65%	3.2	Gage
205	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630822
4.4.2.21	Switch Board Assembly				NEXT HIGHER ASSEMBLY 9390600
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of contacts		0.40%	3.2	Gage
102	Location of contacts in line with two locating holes in operating position (6 places)		0.40%	3.2	Gage
103	Contact force at operating position (6 places)		0.40%	3.2	Gage/Test
Minor					
201	Improper copper wire		0.65%	3.2	Visual
202	Adhesive in area of contact bend or operating area		0.65%	3.2	Visual
203	Evidence of solder bridging prior to adhesive application		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 9390601
4.4.2.22	Plate, Center		NEXT HIGHER ASSEMBLY 9390600

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of pivot holes (4 places)		0.40%	3.2	Gage
102	Location of pivot holes (4 places)		0.40%	3.2	Gage
103	Diameter of base dowel holes (2 places)		0.40%	3.2	Gage
104	Location of base dowel holes (2 places)		0.40%	3.2	Gage
105	Diameter of spacer dowel holes (4 places)		0.40%	3.2	Gage
106	Location of spacer dowel holes (4 places)		0.40%	3.2	Gage
107	Pitch diameter of switch board assy. holes (4 places)		0.40%	3.2	Gage
108	Location of switch board assy. threaded holes (4 places)		0.40%	3.2	Gage
109	Diameter of switch holes (2 places)		0.40%	3.2	Gage
110	Location of switch holes (2 places)		0.40%	3.2	Gage
111	Diameter of escapement holes (2 places)		0.40%	3.2	Gage
112	Width of escapement slots (2 places)		0.40%	3.2	Gage
113	Location of escapement holes (2 places)		0.40%	3.2	Gage
114	Diameter of escapement slots (2 places)		0.40%	3.2	Gage
115	Diameter of spacer mtg. holes (4 places)		0.40%	3.2	Gage
116	Location of spacer mtg. holes (4 places)		0.40%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER 9390601
4.4.2.22 (cont'd)	Plate, Center		NEXT HIGHER ASSEMBLY 9390600

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
117	Pitch diameter of base mtg. holes (2 places)		0.40%	3.2	Gage
118	Location of base mtg. holes (2 places)		0.40%	3.2	Gage
119	Width of switch slot (2 places)		0.40%	3.2	Gage
120	Location of switch slot (2 places)		0.40%	3.2	Gage
121	Diameter of bottom dowel pin holes (2 places)		0.40%	3.2	Gage
122	Location of bottom dowel pin holes (2 places)		0.40%	3.2	Gage
Minor					
201	Protective finish improper, incomplete or missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390600
4.4.2.23	Center Plate Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Delay escapement loose or not against center plate		0.40%	3.2	Visual/Manual
102	Mounting screws for switch board loose or not against board		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of distortion or damage to electrical contacts		0.65%	3.2	Visual
202	Evidence of excess adhesive		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390612
4.4.2.24	Housing "G" Weight Latch				NEXT HIGHER ASSEMBLY 12630813
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of mounting holes		0.40%	3.2	Gage
102	Location of latch hole		0.40%	3.2	Gage
103	Diameter of locating hole		0.40%	3.2	Gage
104	Location of locating hole		0.40%	3.2	Gage
105	Width of locating slot		0.40%	3.2	Gage
106	Location of locating slot		0.40%	3.2	Gage
107	Diameter of latch hole		0.40%	3.2	Gage
108	Location of slot for latch pin		0.40%	3.2	Gage
109	Width of slot		0.40%	3.2	Gage
110	Surface finish in slot and large hole		0.65%	3.2	Visual
Minor					
201	Radius around edge of slot missing (intersection with latch hole)		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12630814
4.4.2.25	Latch, "G" Weight		NEXT HIGHER ASSEMBLY 12630813

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, min.		0.40%	3.2	Gage
102	Inside diameter, min.		0.40%	3.2	Gage
103	Location of small hole		0.40%	3.2	Gage
104	Pitch diameter		0.40%	3.2	Gage
105	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
106	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Radius missing on solid end		0.65%	3.2	Visual
202	Coating missing, incomplete or improper		0.65%	3.4	Visual
203	Surface finish, outside diameter		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	Pin, "G" Weight Latch				12630815
					NEXT HIGHER ASSEMBLY 12630813
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large outside diameter, max.		0.40%	3.2	Gage
102	Pitch diameter, min.		0.40%	3.2	Gage
103	Length		0.40%	3.2	Gage
104	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
105	Coating, adhesive	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Undercut missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0	3.4	Visual

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MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630816
4.4.2.27	Spring, "G" Weight Latch				NEXT HIGHER ASSEMBLY 12630813
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Load at compressed height		0.40%	3.2	Gage
Minor					
201	Protective finish missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630813
4.4.2.28	"G" Weight Latch Housing Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Latch does not move freely		0.40%	3.2	Visual
102	"G" weight latch pin loose		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630806
4.4.2.29	Plate, Shorting, Detonator				NEXT HIGHER ASSEMBLY 12630804
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of mounting holes		0.40%	3.2	Gage
102	Diameter of mounting holes, max.		0.40%	3.2	Gage
103	Diameter of holes for terminal pins, max.		0.40%	3.2	Gage
104	Thickness of plate		0.40%	3.2	Gage
105	Contacts not flush or above		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630805
4.4.2.30	Pin, Terminal				NEXT HIGHER ASSEMBLY 12630804
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large diameter (swage end)		0.40%	3.2	Gage
102	Length (swage end)		0.40%	3.2	Gage
Minor					
201	Finish missing, incomplete or improper		0.65%	3.2	
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630804
4.4.2.31	Shorting Plate, Detonator Assembly				NEXT HIGHER ASSEMBLY 12630803
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460 (2 places)		0.40%	3.2	Visual
Minor					
201	Pin inserted from wrong direction		0.65%	3.2	Visual
202	Pin not swaged (2 places)		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630803
4.4.2.32	Resistor Shorting Plate Detonator Assembly				NEXT HIGHER ASSEMBLY 12630802
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460		0.40%	3.2	Visual
102	Incorrect resistor		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.33	Housing, Detonator				12630807
					NEXT HIGHER ASSEMBLY 12630802
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole for detonator		0.40%	3.2	Gage
102	Depth of hole for detonator		0.40%	3.2	Gage
103	Location of hole for detonator		0.40%	3.2	Gage
104	Location of mounting holes		0.40%	3.2	Gage
105	Location of shorting plate mounting holes (max)		0.40%	3.2	Gage
106	Diameter of shorting plate mounting holes (2 places)		0.40%	3.2	Gage
Minor					
201	Sharp edges at intersection of detonator hole and wire feed through hole		0.65%	3.4	Visual
202	White "S" missing from green field		0.65%	3.2	Visual
203	Paint incorrect or improperly located		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.34	Resistor Shorting Plate Detonator Housing Assembly		1 1		12630802
					NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pin not flush or below (2 places)		0.40%	3.2	Visual
Minor					
201	Excessive adhesive		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390589
4.4.2.35	Spring, "G" Weight				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, min.		0.40%	3.2	Gage
102	Load at 1.300 in.		0.40%	3.2	Gage
103	Load at .575 in.		0.40%	3.2	Gage
Minor					
201	End open		0.65%	3.2	Visual
202	Protective finish		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9392412
4.4.2.36	"G" Weight				NEXT HIGHER ASSEMBLY 9392414
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height		0.40%	3.2	Gage
102	Width		0.40%	3.2	Gage
103	Weight		0.40%	3.2	Scale
104	Height at bushing hole (2 places)		0.40%	3.2	Gage
105	Depth of bushing hole (min) (2 places)		0.40%	3.2	Gage
106	Diameter of bushing hole (max) (2 places)		0.40%	3.2	Gage
107	Location of bushing hole (2 places)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9392413
4.4.2.37	Bushing				NEXT HIGHER ASSEMBLY 9392414
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large outside diameter, max.		0.40%	3.2	Gage
102	Length, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9392414
4.4.2.38	"G" Weight Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of holes (2 places, after axial load test)		0.40%	3.2	Gage
102	Location of holes		0.40%	3.2	Gage
103	Bushing axial load		0.40%	3.2.3	Test
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.39	Weight, Rotor				12630811
					NEXT HIGHER ASSEMBLY
					12630810/9392416
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Weight		0.65%	3.2	Scale
202	Diameter		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.40	Plate, Shorting				12630812
					NEXT HIGHER ASSEMBLY 12630810
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of mounting holes		0.40%	3.2	Gage
102	Diameter of countersunk holes, max. (2 places)		0.40%	3.2	Gage
103	Thickness of plate		0.40%	3.2	Gage
104	Contacts not flush or above		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.41	Rotor, Detonator		9392415
			NEXT HIGHER ASSEMBLY 1263081O

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of pivots (2 places)		0.40%	3.2	Gage
102	Shoulder to shoulder distance of mid-diameter shaft and pivot shaft		0.40%	3.2	Gage
103	Tooth-to-tooth composite tolerance		0.40%	3.2	Gage
104	Surface finish on pivot diameters (2 places)		0.40%	3.2	Gage
105	Location of holes for detonator housing		0.40%	3.2	Gage
106	Pitch diameter of threads for detonator housing mounting (2 places)		0.40%	3.2	Gage
107	Surface finish on shoulder (2 places)		0.40%	3.2	Gage
108	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
109	Coating, adhesion	1	Coupon	3.1.2	Visual/4.5.6.4
Minor					
201	Location of screw thread hole in side (2 places)		0.65%	3.2	Gage
202	Pitch diameter (max) of screw threads in side (2 places)		0.65%	3.2	Gage
203	Location of hole for pin		0.65%	3.2	Gage
204	Diameter of hole for pin (max)		0.65%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER 9392415
4.4.2.41 (cont'd)	Rotor, Detonator		NEXT HIGHER ASSEMBLY 1263081 0

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205	Total composite tolerance		0.65%	3.2	Gage
206	End of gear teeth to centerline (2 places)		0.65%	3.2	Gage
207	Width of slots (2 places)		0.65%	3.2	Gage
208	Location of slots (2 places)		0.65%	3.2	Gage
209	Presence of timing mark		0.65%	3.2	Visual
210	Pitch diameter of threads for detonator housing mounting (2 places)		0.65%	3.2	Gage
211	Location of hole for lead weight		0.65%	3.2	Gage
212	Finish missing or improper		0.65%	3.2	Visual
213	Coating missing, incomplete or improper		0.65%	3.4	Visual
214	Improper color or location of paint		0.65%	3.4	Visual
215	Black "S" missing on red field		0.65%	3.4	Visual
216	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.42	Detonator Rotor Subassembly				12630810
					NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight		0.40%	3.2	Scale
102	Height of straight pin, max.		0.40%	3.2	Gage
103	Shorting plate not against rotor		0.40%	3.2	Visual
Minor					
201	Screws not seated		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390610
4.4.2.43	Actuator, Latch				NEXT HIGHER ASSEMBLY 9390608
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole for solenoid rotor shaft (min)		0.40%	3.2	Gage
102	Width of narrow arm (forward of offset)		0.40%	3.2	Gage
103	Length of narrow arm		0.40%	3.2	Gage
104	Location of offset from hole		0.40%	3.2	Gage
105	Width of offset		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.44	Latch Actuator Solenoid Assembly Solenoid, Rotary		12630826/9390608
			NEXT HIGHER ASSEMBLY 9390606

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance to latch actuator assembly from bottom of solenoid		0.40%	3.2	Gage
102	Location of latch actuator assembly with mounting studs		0.40%	3.2	Gage
103	Spring pin missing		0.40%	3.2	Visual
104	Stakes missing (3 places)		0.40%	3.2	Visual
	SEQUENCE REQUIRED (105, 106, 107)				
105	Output torque with 2 oz. weight @-40°C		100%	3.2	Test
106	Output torque with 3 oz. weight @-40°C		100%	3.2	Test
107	Output torque with 2.4 oz. weight @ 65.6°C		100%	3.2	Test
108	Parallelism		0.40%	3.2	Gage
109	Diameter of rotating shaft		0.40%	3.2	Gage
110	Spring torque @ 2° rotation		0.40%	3.2	Test
111	Spring torque @ 43° rotation, max.		0.40%	3.2	Test
112	Rotation		0.40%	3.2	Gage
113	Stroke direction		0.40%	3.2	Manual
114	Resistance @ 20°C		0.40%	3.2	Test
115	Axial shaft movement, max.		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 1263082 6/9390608
4.4.2.44	Latch Actuator Solenoid Assembly/ Solenoid, Rotary				NEXT HIGHER ASSEMBLY 9390606
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Latch actuator loose on shaft		0.65%	3.2	Manual
202	Wire lead length incorrect		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390607
4.4.2.45	Plate, Top				NEXT HIGHER ASSEMBLY 9390606
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of dowel hole		0.40%	3.2	Gage
102	Location of dowel holes		0.40%	3.2	Gage
103	Width of dowel slot		0.40%	3.2	Gage
104	Location of dowel slot		0.40%	3.2	Gage
105	Diameter of solenoid holes (4 places)		0.40%	3.2	Gage
106	Location of solenoid holes (4 places)		0.40%	3.2	Gage
107	Diameter of corner mtg holes (4 places)		0.40%	3.2	Gage
108	Location of corner mtg holes (4 places)		0.40%	3.2	Gage
Minor					
201	Profile of cut-out		0.65%	3.2	Gage
202	Evidence of poor Workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390606
4.4.2.46	Top Plate Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solenoid orientation improper		0.40%	3.2	Visual
Minor					
201	Lock nut missing or loose (4 places)		0.65%	3.2	Visual/Manual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390603
4.4.2.47	Tee, Spacer				NEXT HIGHER ASSEMBLY 9390692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Perpendicularity of ends		0.40%	3.2	Gage
102	Location of locating hole (4 places)		0.40%	3.2	Gage
103	Diameter of locating hole (4 places)		0.40%	3.2	Gage
104	Pitch diameter of threaded hole, max (6 places)		0.40%	3.2	Gage
105	Location of threaded hole (6 places)		0.40%	3.2	Gage
106	Width of lead housing slot		0.40%	3.2	Gage
107	Location of lead housing slot		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.40%	3.4	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390602
4.4.2.48	Tee Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pin missing (4 places)		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390604
4.4.2.49	Shafting				NEXT HIGHER ASSEMBLY 9390605
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Straightness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390605
4.4.2.50	Rail				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter, min		0.40%	3.2	Gage
102	Length, min		0.40%	3.2	Gage
103	Surface finish		0.40%	3.2	Gage
Minor					
201	Protective finish missing, incomplete or improper		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390596
4.4.2.51	Switch, Lever				NEXT HIGHER ASSEMBLY 9390595
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Operating lever not free		0.40%	3.2	Manual
102	Operating position of lever		0.40%	3.2	Gage
Minor					
201	Lever contact surface damage on radius		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.52	Lever Switch Assembly				9390595
					NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.53	Pin, Rotor Latch		9390599
			NEXT HIGHER ASSEMBLY 9390597

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Small diameter, min		0.40%	3.2	Gage
102	Surface finish on tapered diameter and spherical radius		0.40%	3.2	Gage
103	Length from shoulder to spherical radius		0.40%	3.2	Gage
104	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
105	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390598
4.4.2.54	Latch, Rotor				NEXT HIGHER ASSEMBLY 9390597
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of parallel slots (2 places)		0.40%	3.2	Gage
102	Distance between bottoms of parallel slots		0.40%	3.2	Gage
103	Pin hole diameter		0.40%	3.2	Gage
104	Location of cutout for switch, lever		0.40%	3.2	Gage
105	Location of rectangular cutout for actuator latch		0.40%	3.2	Gage
106	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
107	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
108	Thickness in rectangular cutout		0.40%	3.2	Gage
109	Location of pin hole		0.40%	3.2	Gage
110	Pitch diameter of threaded hole		0.40%	3.2	Gage
111	Surface finish in parallel slots		0.40%	3.2	Visual
Minor					
201	Coating missing, incomplete or improper		0.65%	3.4	Visual
202	Radii in rectangular cutout missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390597
4.4.2.55	Rotor Latch Assembly				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance from rotor latch to end of pin, max		0.40%	3.2	Gage
102	Damage on spherical radius		0.40%	3.2	Visual
Minor					
201	Protective finish missing, incomplete or improper (except holes)		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390593
4.4.2.56	Detent, Rotor Latch				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		0.40%	3.2	Gage
102	Length from hole to narrow end		0.40%	3.2	Gage
103	Profile		0.40%	3.2	Gage
104	Finish of step		0.40%	3.2	Visual
105	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
106	Coating adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Thickness		0.65%	3.2	Gage
202	Coating missing, incomplete or improper		0.65%	3.4	Visual
203	Evidence of poor workmanship		1.0%	3.4	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.57	Bearing, Rotor Latch Detent				12630820
					NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of bearing surface		0.40%	3.2	Gage
102	Distance from retainer ring groove to shoulder		0.40%	3.2	Gage
103	Depth of slot across shoulder, min		0.40%	3.2	Gage
104	Surface finish on bearing surface		0.40%	3.2	Visual
105	Shoulder to groove thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 1263081 7
4.4.2.58	Anchor, Rotor Latch Spring				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Large diameter, min		0.65%	3.2	Gage
202	Small diameter, max		0.65%	3.2	Gage
203	Protective finish missing, incomplete or improper		0.65%	3.2	Visual
204	Location of notch		0.65%	3.2	Gage
205	Depth of notch		0.65%	3.2	Gage
206	Width of notch		0.65%	3.2	Gage
207	Location of slot (2 places)		0.65%	3.2	Gage
208	Depth of slot (2 places)		0.65%	3.2	Gage
209	Width of slot (2 places)		0.65%	3.2	Gage
210	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630818
4.4.2.59	Spring, Rotor Latch Detent				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside radius		0.40%	3.2	Gage
102	Inside radius		0.40%	3.2	Gage
103	Improper configuration		0.40%	3.2	Visual
104	Width of opening for detent, min		0.40%	3.2	Gage
105	Length from end to center of coil		0.40%	3.2	Gage
Minor					
201	Protective finish missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630819
4.4.2.60	Spring, Rotor Latch				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Inside diameter of loop, min		0.65%	3.2	Gage
202	Angle between legs		0.65%	3.2	Gage
203	Diameter of wire		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER 9390594
4.4.2.61	Plate, Outer		NEXT HIGHER ASSEMBLY 9390592

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of pivot holes (2 places)		0.40%	3.2	Gage
102	Location of rail holes (4 places)		0.40%	3.2	Gage
103	Diameter of pivot holes (2 places)		0.40%	3.2	Gage
104	Diameter of rail holes (4 places)		0.40%	3.2	Gage
105	Flatness of bottom surface		0.40%	3.2	Gage
106	Location of hole for rotor latch detent		0.40%	3.2	Gage
107	Diameter of rotor latch detent hole		0.40%	3.2	Gage
108	Length of slot for latch		0.40%	3.2	Gage
109	Width of slot for latch, entire length		0.40%	3.2	Gage

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9390594
4.4.2.61 (cont'd)	Plate, Outer				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
110	Surface finish in pivot holes		0.40%	3.2	Gage
111	Flatness rotor surface		0.40%	3.2	Gage
112	Diameter of spacer dowel holes (2 places)		0.40%	3.2	Gage
113	Location of spacer dowel holes (2 places)		0.40%	3.2	Gage
114	Pitch diameter of base plate mtg. holes (2 places)		0.40%	3.2	Gage
115	Location of base plate mtg. holes (2 places)		0.40%	3.2	Gage
116	Diameter of spacer mtg. holes (3 places)		0.40%	3.2	Gage
117	Location of spacer mtg. holes (3 places)		0.40%	3.2	Gage
118	Diameter of "G" wgt. latch dowel holes (2 places)		0.40%	3.2	Gage
119	Location of "G" wgt. latch dowel holes (2 places)		0.40%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9390594
4.4.2.61 (cont'd)	Plate, Outer				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
120	Pitch diameter of "G" wgt. latch mtg. holes		0.40%	3.2	Gage
121	Location of "G" wgt. latch mtg. holes		0.40%	3.2	Gage
122	Pitch diameter of switch mtg. hole		0.40%	3.2	Gage
123	Location of switch mtg. hole		0.40%	3.2	Gage
124	Width of switch mtg. slot		0.40%	3.2	Gage
125	Location of switch mtg. slot		0.40%	3.2	Gage
126	Height of switch mtg. slot		0.40%	3.2	Gage
127	Thickness for lower "G" wgt. surface		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630847
4.4.2.62	Adjuster, Nut Switch				NEXT HIGHER ASSEMBLY 9390592
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter		0.40%	3.2	Gage
102	Step, max. (2 places)		0.40%	3.2	Gage
Minor					
201	Thickness, max.		0.65%	3.2	Gage
202	Width, min.		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390592
4.4.2.63	Outer Plate Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Rotor latch detent does not move freely		0.40%	3.2	Manual
102	Retaining ring on rotor latch detent missing or not in groove		0.40%	3.2	Visual
103	Mounting block not against plate or loose		0.40%	3.2	Visual
104	Lever switch orientation incorrect		0.40%	3.2	Visual
105	Lever switch assembly screws loose		0.40%	3.2	Visual
106	Rotor latch assembly does not move freely		0.40%	3.2	Manual
107	Rotor latch spring not in notch (2 places)		0.40%	3.2	Visual
108	Spacer not present, if necessary, to prevent override of rotor latch by rotor latch detent		0.40%	3.2	Manual
Minor					
201	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390613
4.4.2.64	Screw, Self-sealing				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Head, diameter		0.65%	3.2	Gage
202	Head, angle		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Pitch diameter, min.		0.65%	3.2	Gage
205	"O" ring not present		0.65%	3.2	Visual
206	Evidence of poor workmanship		1.0%	3.4	Visual

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MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.65	Board, Stiffener				9390611
					NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of mounting holes (4 places)		0.40%	3.2	Gage
102	Location of mounting holes (4 places)		0.40%	3.2	Gage
103	Diameter of holes (12 places)		0.40%	3.2	Gage
104	Location of holes (12 places)		0.40%	3.2	Gage
105	Location of cutout in side (2 places)		0.40%	3.2	Gage
Minor					
201	Edge chamfer missing		0.65%	3.2	Visual
202	Radius missing on corners (4 places)		0.65%	3.2	Visual
203	Profile of cutouts (2 places)		0.65%	3.2	Gage
204	Diameter of solenoid mtg holes (4 places)		0.65%	3.2	Gage
205	Location of solenoid mtg holes (4 places)		0.65%	3.2	Gage
206	Evidence of poor workmanship		1.0%	3.4	Visual

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MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390591
4.4.2.66	Plate, Base				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of four warhead large thru holes		0.40%	3.2	Gage
102	Location of four warhead large thru holes		0.40%	3.2	Gage
103	Location of cover hole (8 places)		0.40%	3.2	Gage
104	Diameter of cover hole (8 places)		0.40%	3.2	Gage
105	Location of warhead orientation notch		0.40%	3.2	Gage
106	Depth of lead output holes (2 places)		0.40%	3.2	Gage
107	Flatness		0.40%	3.2	Gage
108	Outside dimension (2 places)		0.40%	3.2	Gage
109	Diameter of "A" hole (7 places)		0.40%	3.2	Gage
110	Location of "A" hole (7 places)		0.40%	3.2	Gage
Minor					
201	Radius missing, outside corners		0.65%	3.2	Visual
202	Countersink missing, improper side or breaking through		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9390615
4.4.2.67	Gear, Escapement		NEXT HIGHER ASSEMBLY 9392416

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total tooth profile (not root diameter or top blend radius)		0.40%	3.2	Gage
102	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
103	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Location of two mounting holes		0.65%	3.2	Gage
202	Diameter of mounting holes		0.65%	3.2	Gage
203	Finish of gear teeth		0.65%	3.2	Visual
204	Coating missing, incomplete or improper		0.65%	3.4	Visual
205	Evidence of poor workmanship		1.0%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390616
4.4.2.68	Rotor, Balance				NEXT HIGHER ASSEMBLY 9392416
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of pivots (2 places)		0.40%	3.2	Gage
102	Shoulder to shoulder distance of mid-diameter		0.40%	3.2	Gage
103	Total composite tolerance		0.40%	3.2	Gage
104	Tooth-to-tooth composite tolerance		0.40%	3.2	Gage
105	Finish of pivot (2 places)		0.40%	3.2	Gage
106	Coating, thickness	1	Coupon	3.1.2	Gage/4.5.6.3
107	Coating, adhesion	1	Coupon	3.1.2	Test/4.5.6.4
Minor					
201	Location of threaded holes (2 places)		0.65%	3.2	Gage
202	Pitch diameter of screw threads, max (2 places)		0.65%	3.2	Gage
203	Presence of timing mark		0.65%	3.2	Visual
204	Coating missing, imcomplete or improper		0.65%	3.4	Visual
205	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9392416
4.4.2.69	Balance Rotor Assembly				NEXT HIGHER ASSEMBLY 9390590
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Escapement gear not against rotor		0.65%	3.2	Visual/Manual
202	Escapement gear loose		0.65%	3.2	Visual/Manual
203	Pan head screw not seated (2 places)		0.65%	3.2	Visual/Manual
204	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9390590
4.4.2.70	Mechanical Assembly				NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Timing marks not aligned properly		0.40%	3.2	Visual
102	Detonator rotor and balance rotor gear teeth do not engage when rotated		0.40%	3.2	Visual/Manual
103	Balance rotor assembly and escapement gear teeth and escapement assembly do not engage when rotated		0.40%	3.2	Visual/Manual
104	Top plate to center plate screws missing or not fully seated		0.40%	3.2	Visual
105	Center plate to top plate locating pins above flush		0.40%	3.2	Visual
106	Solenoid actuator not engaged in rotor latch (2 places)		0.40%	3.2	Visual
107	Rotor latch assembly and rotor latch detent do not move freely upon manual operation of solenoid actuator (2 places)		0.40%	3.2	Visual/Manual
108	"G" weight springs missing (2 places)		0.40%	3.2	Visual
109	"G" weight latch detent springs missing (2 places)		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9390590
4.4.2.70	Mechanical Assembly (Continued)				NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Evidence of scratched or damaged rails		0.65%	3.2	Visual
202	Base plate screws missing or not fully seated (6 places)		0.65%	3.2	Visual
203	Board stiffener screws missing (4 places)		0.65%	3.2	Visual
204	Board stiffener washers missing (3 places)		0.65%	3.2	Visual
205	Board stiffener insulated washer missing (1 place)		0.65%	3.2	Visual
206	Serial number missing		0.65%	3.2	Visual
207	Board stiffener not flush with top plate		0.65%	3.2	Visual
208	Evidence of poor workmanship		1.0%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48686 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9390590
4.4.2.71	Mechanical Assembly (Electrical/ Mechanical Operation)		NEXT HIGHER ASSEMBLY 12630801

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity (Safe and Locked)		100%	3.3.1	Test
102	Isolation resistance (Safe and Locked)		100%	3.3.2	Test
103	Intent-to-launch (ITL)		100%	3.3.3	Test
104	Continuity (Safe and Unlocked)		100%	3.3.1	Test
105	Detonator rotor arming		100%	3.3.4	4.5.5/Test
106	Continuity (Armed and Locked)		100%	3.3.1	Test
107	Isolation resistance (Armed and Locked)		100%	3.3.2	Test
108	Return to safe		100%	3.3.6	Test
109	Non-arming		100%	3.3.5	4.5.6/Test

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Molded components. At the start of production from a new mold or after adjustment or repair to a mold, consecutively produced units from each mold shall be dimensionally examined until four (4) consecutive units meet all applicable drawing requirements. After adjustment or repair, those dimensions affected by the adjustment or repair shall be examined. As a control of each cavity during production, three (3) parts from each cavity shall be inspected for all applicable drawing requirements after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found, that portion of production since the last control check shall be inspected for each separate type of defect.

4.5.2 Numerical control parts. In place of normal sampling associated with classification of defects, consecutively produced parts shall be dimensionally examined until four (4) consecutive parts meet all applicable drawing requirements when either a tape is prepared or changed. During production, one piece part from each four (4) hours production shall be inspected for the location of the first and last hole drilled of each size. If the sample piece part fails, all other items in that production lot shall be screened for each type of defect.

4.5.3 Stamped parts. In place of the normal sampling associated with the Classification of Defects, a sample of at least ten (10) parts shall be dimensionaly inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

4.5.4 Procedure for detonator rotor arming test.

4.5.4.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. Mechanical Assembly Safe and Unlocked

b. S2 (L) E3 to E4 Closed

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c. S2 (R) E3 to E4 Closed

d. S1 (L) E1 to E2 Closed

e. S1 (R) E1 to E2 Closed

4.5.4.2 Dynamic conditions. The test equipment shall retain the Mechanical Assembly in a position such that its -Z axis initially is subjected to 0 ± 0.2g's while the X or Y axis is being subjected to an acceleration level of 11.9 ± 0.2g's ("Out-of-Line" See figure 2a). When the test acceleration level has been attained, the test equipment shall smoothly transition the -Z axis of the Mechanical Assembly 90 ± 5°, within a period of time not greater than 300msec, to the condition where the Mechanical Assembly -Z axis is now experiencing the test acceleration level ("In-Line" See figure 2b). The Mechanical Assembly Device shall be positioned such that the axis of rotation of the test equipment is through the center of gravity (CG). Record the times as specified in 3.3.4.1.

4.5.5 Procedure for non-arming test.

4.5.5.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. Mechanical Assembly Safe & Unlocked

b. S2 (L) E3 to E4 Closed

c. S2 (R) E3 to E4 Closed

d. S1 (L) E1 to E2 Closed

e. S1 (R) E1 to E2 Closed

4.5.5.2 Dynamic conditions. The test equipment shall preposition the -Z axis of the Mechanical Assembly (see figure 1) in the g field (see figure 2b) and subject the -Z axis to a constant 4.0 + 0.5g acceleration. The Mechanical Assembly shall not arm as specified in 3.3.5. At the completion of this test the test equipment shall increase the acceleration on the -Z axis to a level that will allow both detonators to Arm and Lock. The test equipment shall not exceed a g level of 12.1g's.

4.5.6 Coupons for polymer coating. In order to perform adhesion testing and dimensional analysis, a test coupon shall be supplied with each lot of parts to be processed with Coating, Polymer Film Lubricant. A lot of parts is defined as parts of the same base metal to be coated under the same conditions and by means of the same coating process.

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4.5.6.1 Types of coupon. The dimensions by type shall not deviate from any dimension by more than 25%.

4.5.6.1.1 Rectangular. The dimensions shall be 1.5" x 0.5" x 0.04.

4.5.6.1.2 Cylindrical. The dimensions shall be 1.5" long with a 0.125" diameter.

4.5.6.2 Codes and material.

4.5.6.2.1 Codes. Each test coupon shall be marked with two identifying codes. The material code is as specified in Table IV.

Table IV. Material codes

Code	Material Description	Material Spec.
S1	Carbon steel, 1137 or 1142	ASTM-A108
S2	Steel, wire, stainless cold finished type 416 or 416 5e condition T or H	ASTM-A581
S3	Stainless and heat resisting chromium nickel steel, sheet or strip, type 301 or 302	ASTM-A167
S4	Steel, sheet or strip, cold rolled 1035 to 1075	QQ-S-700
C1	Brass, strip, copper alloy 260, 268 or composition II, half hard	QQ-B-613
C2	Copper, beryllium, strip copper alloy C17200, temper TH02	ASTM-B194

4.5.6.2.1.1 Code. The second identifying code is left to the discretion of the contractor. It must be adequate to associate the test coupon to the represented lot of parts. The two identifying codes are to be stamped on the coupons away from the center of the coupon so as no to interfere with the surface of interest in the adhesion test. Deformation of the surface in the area of the stamping is permissible.

4.5.6.2.2 Material. Each coupon shall be of the identical material and in the same condition as the part(s) it represents. This is to include any treatments such as passivation or heat treatment to which the parts are subjected to before the application of the polymer film coating.

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4.5.6.3 Thickness inspection.

4.5.6.3.1 Microscopic method. The coating thickness shall be determined in accordance with ASTM-B487. A minimum of five (5) measurements are to be taken and the average of these measurements will determine the coating thickness. Otherwise and instrument method may be employed.

4.5.6.3.2 Instrument method. The instrument, its calibration, and its operation shall be such that the coating thickness will be determined within ± 10 percent of its true thickness.

4.5.6.4 Adhesion test. Rectangular test coupons will be bent 180 ± 10 degrees (relaxation to a lesser angle after initially reaching 180 ± 10 degrees is permitted) about a mandrel with a diameter no greater than four (4) times the thickness of the test coupon. Cylindrical test coupons are to be bent 90 ± 10 degrees (relaxation to a lesser angle after initially reaching 90 ± 10 degrees is permitted) about a mandrel with a diameter no greater than four (4) times the diameter of the test coupon. The coupon is then to be examined at four (4) power magnification in the outer radius of the bend for flaking or separation of the coating from the base metal. The criteria for acceptance shall be that the coupon exhibits no evidence of flaking or separation of the coating from the coupon.

If the test coupon fractures during the bending process before the desired angle is reached, an alternate inspection is to be employed. First the area of the fracture is to be inspected for evidence of flaking or separation of the coating. In addition, a sharp instrument (Exacto Slimstyle Knife Catalog No. H3-3201, Blade No. H3-416, or equivalent) shall be used to scratch a cross hatched pattern on the larger of the two vertical lines and two horizontal lines extending the full length and width of the test coupon. The pattern is to consist of two vertical lines and two horizontal lines extending the full length and width of the coupon piece. The scratched lines are to be sufficiently deep to reach the base metal. The lines should appear as continuous lines with no evidence of flaking or separation of the coating.

Failure of the test coupon to meet either the thickness or adhesion criterion shall be cause for rejection of all parts which the test coupon represents.

4.5.7 Tolerances.

4.5.7.1 Environmental (mechanical). Tolerances shall be those specified in MIL-STD-810.

4.5.7.2 Measurements (electronic). Tolerances specified herein are absolute with no allowance made for equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

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5. PACKAGING - Not applicable

6. NOTES.

6.1 Intended use. This specification is intended to be used in conjunction with MIL-S-48687, Safety and Arming Device, GM: M143E1 Loading, Assembling and Packing For. In the event of conflict, MIL-S-48687 shall take precedence.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-I (D), Dover, New Jersey 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Key word listing.

Safety and Arming Device
Mechanical parts

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1336-A530)

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